

Dabin Kim

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Profile

I am an ML/DL researcher focused on generative models for music and audio, particularly *fine-grained offline editing* and *real-time online interaction*. My current research explores full-duplex systems that listen and generate concurrently, enabling responsive musical co-creation with users.

Affiliation: Music and Audio Computing (MAC) Lab, KAIST

Research Interests: Controllable Music Generation, Real-time Musical Interaction



Education

Korea Advanced Institute of Science and Technology (KAIST), Daejeon Ph.D. in Music and Audio Computing (MAC) Lab	Mar 2026 – Present
Korea Advanced Institute of Science and Technology (KAIST), Daejeon M.S. in Music and Audio Computing (MAC) Lab	Aug 2023 – Aug 2025
Sogang University, Seoul B.S. in Art & Technology (Double Major: Big Data Science)	Feb 2017 – Feb 2023

Experience

Sony Computer Science Laboratories (Sony CSL) Assistant Researcher	Jul 2025 – Oct 2025
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Publications

- [1] **AdaTT: Text-Guided Instrument Timbre Transfer with Target-Adaptive Structural Control**
Dabin Kim, Junwon Lee, Juhan Nam
27th Annual Conference of the International Speech Communication Association (INTERSPEECH), 2026
[\[Paper Link\]](#) [\[Demo Link\]](#)
- Proposed a target-adaptive control framework for timbre transfer that preserves score-level content while improving target timbral fidelity.
 - Introduced a semi-automatic data pipeline for learning instrument-specific expressive transformations across diverse source–target scenarios.
- [2] **DDSP-Based Neural Vehicle Sound Synthesis from Driving Signals**
Minsuk Choi, Dabin Kim, Daehun Song, Juhan Nam
6th AES International Conference on Automotive Audio, 2026
[\[Paper Link\]](#) [\[Demo Link\]](#)
- Proposed a real-time neural vehicle sound synthesis framework based on Differentiable Digital Signal Processing (DDSP).

[3] **Video-Foley: Two-Stage Video-To-Sound Generation via Temporal Event Condition for Foley Sound**

Junwon Lee, Jaekwon Im, **Dabin Kim**, Juhan Nam

IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2025

[\[Paper Link\]](#) [\[Demo Link\]](#) [\[Code Link\]](#)

- Contributed to video-conditioned intensity modeling and qualitative evaluation for temporally synchronized Foley generation.

[4] **Pitch-ControlNet: Continuous Pitch Control for Monophonic Instrument Sound Generation**

Dabin Kim*, Junwon Lee*, Minseo Kim*, Juhan Nam (**equal contribution*)

Late-Breaking/Demo (LBD), 25th International Society for Music Information Retrieval Conference (ISMIR), 2024

[\[Paper Link\]](#)

- Proposed continuous pitch control for instrument sound generation using ControlNet.

Selected Presentations

AdaTT: Text-Guided Instrument Timbre Transfer with Target-Adaptive Structural Control

Sep 27 – Oct 1, 2026

INTERSPEECH 2026

International Convention Centre Sydney, Sydney, Australia

AI generative models for controllable music and Foley sound synthesis

May 12, 2026

KOBA 2026 AI Audio/Sound Day

COEX Conference Room, Seoul, South Korea

Skills

Programming Languages: Python

ML/DL Frameworks: PyTorch, PyTorch Lightning

Creative & Audio Tools: Ableton Live, Logic Pro, Max/MSP, SuperCollider